GWO-WOA-AOA: Multistage Hybrid Meta-heuristic Optimization Approach

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***Abstract*— A new hybrid optimization methodology merging the advantages of three distinct meta-heuristic techniques—Grey Wolf Optimization Algorithm (GWO), Whale Optimization Algorithm (WOA), and Arithmetic Optimization Algorithm (AOA)—is introduced in this study. The proposed hybrid framework addresses the optimization process with a three-stage approach: The exploration phase with GWO, the exploitation phase with WOA, and the fine-tuning phase with AOA. The performance of the proposed hybrid method is evaluated on CEC2019 benchmark functions (problems with different difficulties in total of 10). The experimental results show that the proposed hybrid model can achieve faster convergence and superior solution accuracy compared to conventional methods, especially in higher dimensions and problems with multiple optima.**

Keywords*— *Hybrid optimization, meta-heuristic algorithms, GWO, WOA, AOA

I. INTRODUCTION

Optimization techniques are very essential part of today's complex engineering problems. Recently, with the poor capacity of conventional approaches in tackling multidimensional, constrained nonlinear and multimodal problems nature-inspired metaheuristic techniques has grown massively due to the advent [1]. These algorithms, by following the behaviors of living organisms in nature lead to actionable techniques for solving complex optimization problems [2].

One of the most important features of metaheuristic algorithms is their ability to balance exploration and exploitation between an ideal harmony. One is the driven of exploring (to explore different part of space of a problem solution) and the other one is exploitation-so more concentrated searches in promising areas [3]. It is however very rare for a single algorithm to work perfectly both ways. Metaheuristic algorithms are usually good for either exploration or exploitation, most probably [4], [5]. This means hybrid approaches that combine the best of different algorithms have the potential to drastically enhance the optimization performance [6].

Grey Wolf Optimization Algorithm (GWO) is a metaheuristic that follows the social structure and hunting technique from grey wolves [7]. GWO solves the solution space so efficiently by using hierarchical (alpha, beta, delta) wolves. Whale Optimization Algorithm (WOA) is the algorithm that tries to mimic the bubble net hunting strategy of humpback whales and does really good exploitation is an implementation [8]. The Arithmetic Optimization Algorithm (AOA) employs basic mathematical operations (multiplication, division, subtraction and addition) to perform precise searches over the solution space in a novel metaheuristic way [9].

In this paper, a novel hybrid optimization model by combining the merits of three metaheuristic algorithms in details is proposed. The suggested hybrid scheme implements the optimization process as a three-stage framework: GWO in the exploration phase; WOA in the exploitation one, and AOA for the fine-tuning process. That three-phase framework aimed at optimizing global search based on preserving local search quality leads to much better and balanced optimization process.

In this work, the performance evaluation of the proposed GWO-WOA-AOA hybrid technique is stated by using CEC2019 benchmark functions. The CEC2019 test set is composed of 10 benchmark test functions (at various levels of hardness) that are extensively used for testing the performance of optimization techniques. Extensive experimental evaluation prove that the proposed hybrid approach surpasses traditional methods in literature (PSO, DE and ALO). Hybrid algorithm for multivariable and mutiobjective problems converges faster and produces better results.

A. Literature Review

Recently, metaheuristic studies were focused on proposing new techniques in recent years and numerous studies have performed on optimizing the existing algorithms. In this section, a brief literature review of the hybridization of three algorithms implemented in the present work (GWO, WOA and AOA) with other methods.

Various review studies have considered hybridization of GWO with other algorithms. Fatih et al. proposed an hybridization of PSOGWO by combining GWO and Particle Swarm Optimization (PSO) algorithms [10]. Through this hybrid approach, the following more balanced optimization process between the refinement ability of PSO and exploration capability of GWO was achieved. Digra et al suggested an ABCGWO hybrid algorithm by integrating GWO and Artificial Bee Colony (ABC) algorithms [11]. This hybrid approach merges the GWO proficiency for working in promising areas and exploration ability of ABC to deliver better outputs in multi-modal optimization problems.

There are some works about hybrid WOA with other algorithms. WOA and Simulated Annealing (SA) algorithms have been integrated with an integrated technique for feature selection problem from Mafarja and Mirjalili [12]. The SA's solution refinement ability and WOA's ability to explore large space of candidate solution were combined together in this hybrid approach for the better results. Likewise, Chen et al. the Chaotic Whale Optimization Algorithm (CWOA), which is the hybrid algorithm of WOA and Chaos Theory was suggested [13]. The proposed hybrid method tackles the drawback of being trapped in local optima with the help of boosting chaotic maps to WOA for generating diversity. In addition, Ling et al. proposed an integrated approach called LWOA that merges with WOA and Lévy flight strategy [14].

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By using this hybrid approach, the convergence property of WOA and the random search capability of Lévy flight work together to provide the results for global optimization problems more effective.

There are still few studies on the hybridization of AOA with other algorithm due to it as a new algorithm. Do D et al. suggested a hybrid method named AOADE combining AOA and Differential Evolution (DE) algorithms [15]. The hybrid approach of this work combines both, the diversity of mathematical operators from AOA and mutation and crossover strategies of DE in order to solve more complex problems. Yıldız et al. developed a combined algorithm by merging AOA and PSO algorithms that called AOAPSO [16]. This hybrid technique implores the fine-tuning behavior of AOA's operators in mathematical environment with social learning mechanism of PSO for outcomes in higher-dimensional optimization problems.

Literature review reveals that hybrid metaheuristic algorithms performed better than a single algorithm itself [17]. Hybrid approaches of most hybrids are just two algorithms and incremental hybrid approaches with three or more algorithms are very limited. An Integrated Optimization Method: The new integrated algorithm, as introduced in this study combines efficiencies of three other algorithms (GWO, WOA & AOA) and also deploy the optimization process by a three steps approach. Hybrid approach: integrates three essential strengths of GWO's search, WOA's exploitation and AOA's fine-tuning which gives a rational and efficient solution for complex optimization problems.

II. MATERIAL AND METHOD

A. Grey Wolf Optimization Algorithm (GWO)

A nature-inspired metaheuristic optimization technique, the GWO algorithm was brought forth by Mirjalili et al. in 2014 [7]. The GWO algorithm operates by replicating the grey wolves' pack structure and hunting strategies. The algorithm models the social hierarchy in the wolf pack at four levels: alpha (α), beta (β), delta (δ), and omega (ω) wolves. In optimization, solutions are hierarchically ranked: the best as alpha, the second best as beta, and the third best as delta, with all others classified as omega wolves [18]. Grey wolves employ a hunting strategy encompassing three primary phases: tracking and approaching the prey, surrounding and immobilizing it, and finally attacking it [7]. These behaviors are formulated in mathematical terms within the GWO. The following equations express prey encirclement behavior:

$$\vec{D} = \left|\vec{C} \cdot \vec{X}_{p(t)} - \vec{X}(t)\right| \quad (1)$$

$$\vec{X}(t+1) = \vec{X}_{p(t)} - \vec{A} \cdot \vec{D} \quad (2)$$

Where t is the current iteration, $\vec{X}_p$ is the hunt location (best solution), $\vec{X}$ is the current location of the grey wolf, $\vec{A}$ and $\overrightarrow{C}$ are the coefficient vectors. The calculation of these coefficient vectors is performed in the following manner:

$$\vec{A} = 2\vec{a} \cdot \overrightarrow{r_1} - \vec{a} \quad (3)$$

$$\vec{C} = 2 \cdot \overrightarrow{r_2} \quad (4)$$

Where $\overrightarrow{a}$ is a value that decreases linearly from 2 to 0 over the iteration, and $\overrightarrow{r_1}$ and $\overrightarrow{r_2}$ are random vectors in the interval [0, 1].

The omega wolves in the GWO algorithm modify their positions by the alpha, beta, and delta wolves, which have more accurate information about where the prey might optimally be located [19]. This process is expressed by the equations presented below:

$$\vec{D}_\alpha = \left|\vec{C}_1 \cdot \vec{X}_\alpha - \vec{X}\right|, \quad \vec{D}_\beta = \left|\vec{C}_2 \cdot \vec{X}_\beta - \vec{X}\right|, \quad \vec{D}_\delta = \left|\vec{C}_3 \cdot \vec{X}_\delta - \vec{X}\right| \quad (5)$$

$$\vec{X}_1 = \left(\vec{X}_\alpha - \vec{A}_1 \cdot \vec{D}_\alpha\right), \quad \overrightarrow{X}_2 = \left(\vec{X}_\beta - \vec{A}_2 \cdot \vec{D}_\beta\right), \quad \overrightarrow{X}_3 = \left(\vec{X}_\delta - \vec{A}_3 \cdot \vec{D}_\delta\right) \quad (6)$$

$$\vec{X}(t+1) = (\vec{X}_1 + \overrightarrow{X}_2 + \vec{X}_3)/3 \quad (7)$$

Throughout the iterations of the GWO algorithm, the parameter $\vec{a}$ undergoes a linear reduction from 2 to 0:

$$\overrightarrow{a} = 2 - t \cdot (2/T_{max}) \quad (8)$$

Here $T_{\max}$ is the maximum number of iterations. The value of the vector $\vec{A}$ controls how the algorithm balances between exploitation and exploration phases [20]: When |A| < 1, the wolves advance in the direction of the prey (exploitation); when |A| > 1, wolves conduct exploration by moving away from known prey in search of more favorable options. Furthermore, vector C provides random weights of the movement of the prey (C > 1 or C < 1), contributing to the GWO algorithm's effectiveness at avoiding local optimum entrapment [21].

The GWO algorithm proceeds through the following steps: (1) Establish the initial wolf population, (2) Evaluate each wolf's fitness value, (3) Determine the alpha, beta, and delta wolves, (4) Update the parameter a at each iteration, calculate the coefficient vectors A and C for each wolf, update the position of each wolf to the alpha, beta and delta wolves, reevaluate the population and update the alpha, beta and delta wolves, (5) Repeat step 4 until the stopping criterion is met, (6) Return the best solution (the position of the alpha wolf).

B. Whale Optimization Algorithm

A technique that simulates the hunting patterns of humpback whales, the Whale Optimization Algorithm (WOA) stands as one of the nature-based optimization methods [8]. Developed in 2016 by Mirjalili and Lewis, this algorithm mathematically models the hunting strategy of humpback whales by creating a bubble network [22]. WOA is based on three main mechanisms: prey containment, bubble net fishing strategy, and prey search [23]. These mechanisms are represented through the subsequent mathematical formulas: Equations used for prey containment behavior:

$$\vec{D} = \left|\vec{C} \cdot \overrightarrow{X^*} - \vec{X}\right| \quad (9)$$

$$\vec{X}(t+1) = \overrightarrow{X^*} - \vec{A} \cdot \vec{D} \quad (10)$$

In these equations, t is the number of iterations, $\overrightarrow{X^*}$ represents the location of the optimal solution discovered until now, $\vec{X}$ is the current position of the whale [24]. The coefficient vectors $\vec{A}$ and $\vec{C}$, the formula for their computation is presented below:

$$\overrightarrow{A} = 2\overrightarrow{a} \cdot \vec{r} - \vec{a} \quad (11)$$

$$\vec{C} = 2 \cdot \vec{r} \quad (12)$$

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Where $\vec{a}$ decreases from 2 to 0 over iterations and $\vec{r}$ is a randomly generated vector in the range [0,1]. The bubble net fishing strategy is modeled by two different approaches [22]:

1. The shrinking circle approach: In this approach, the value of $\vec{a}$ decreases, the value of $\vec{A}$ also decreases, and equation (10) is used.

2. Spiral update approach:

$$\vec{X}(t+1) = \overrightarrow{D'} \cdot cos(2\pi l) + \overrightarrow{X^*} \quad (13)$$

$\overrightarrow{D'} = \left|\overrightarrow{X^*} - \vec{X}\right|$ indicates how far the whale is from the prey, b is a fixed value, and l is a random number in the range [-1,1] [23].

Mathematical formulations employed during the algorithm's search phase:

$$\vec{D} = |\vec{C} \cdot \vec{X}_{rand} - \vec{X}| \quad (14)$$

$$\vec{X}(t+1) = \vec{X}_{rand} - \vec{A} \cdot \vec{D} \quad (15)$$

Here $\vec{X}_{rand}$ denotes an individual randomly selected from the population [4].

In WOA's operation, a probability value p (usually 0.5) is used to choose between a shrinking circle or a spiral update mechanism. Furthermore, when $|\vec{A}|$ is less than 1, the exploitation process is carried out, and the exploration stage takes place when the magnitude of the vector when $|\vec{A}|$ exceeds 1. [23].

The fundamental procedures of the algorithm are outlined below [8]:

1. Form the starting population
2. Evaluate fitness values and determine the best solution
3. At each iteration:
3.1 Update parameters a, A, and C
3.2 Generate random p values
3.3 If p < 0.5:
3.3.1 If |A| < 1, apply the shrinking circle mechanism
3.3.2 If |A| ⩾ 1, perform exploration by randomly selecting whales
3.4 If p ⩾ 0.5, apply the spiral update mechanism
4. Continue until the stopping criterion is met

C. Arithmetic Optimization Algorithm (AOA)

Inspired by the conduct of mathematical operators, the Arithmetic Optimization Algorithm (AOA) functions as a meta-heuristic approach to optimization. The AOA begins with a randomly created population in its first stage. The procedure for generating the starting population and the ensuing mathematical expression is outlined as follows:

The initial population consists of N solution candidates, and each solution is generated in d-dimensional space using the equation detailed below:

$$X_i = LB + rand(0,1) \cdot (UB - LB), i = 1,2,\ldots,N \quad (16)$$

Where X_i is the i. solution candidate, LB and UB are the lower and upper bounds, respectively, and rand (0,1) is the random number generated in the interval [0,1]. Once the initial population is created, the Mathematical Operator Proportion (MOP) is calculated at each iteration of the algorithm:

$$MOP(t) = 1 - \left(\frac{t}{T}\right)^{\alpha} \quad (17)$$

In equation (17), t denotes the current iteration, T indicates the maximum number of iterations, and α the control parameter. The MOP value regulates the transition amid the algorithm's phases of exploitation and exploration. The algorithm is divided into two phases by comparing the random number r_1 with MOP. In the case of $r_1 > MOP$, the local search phase is active, and the equations presented below are employed:

$$X_i^{t+1} = \frac{X_{best}}{MOP} \cdot \left((UB - LB) \cdot \mu + LB\right), \quad for\ r_3 < 0.5 \quad (18)$$

$$X_i^{t+1} = X_{best} \cdot MOP \cdot \left((UB - LB) \cdot \mu + LB\right), \quad for\ r_3 \geq 0.5 \quad (19)$$

If $r_1 \leq MOP$, the fine-tuning phase is activated:

$$X_i^{t+1} = X_{best} - MOP \cdot \left((UB - LB) \cdot \mu + LB\right), \quad for\ r_2 < 0.5 \quad (20)$$

$$X_i^{t+1} = X_{best} + MOP \cdot ((UB - LB) \cdot \mu + LB), \quad for\ r_2 \geq 0.5 \quad (21)$$

The parameters r_1, r_2, and r_3 are randomly generated values in the interval [0,1]. Equations (18) and (19) provide an efficient exploration of the solution space throughout the local search process, while Equations (20) and (21) improve the ability to converge to the local optimum in the fine-tuning phase.

D. Proposed GWO_WOA_AOA algortihm

The hybrid algorithm combines the good of three various algorithms to provide better high performing and exact solutions for optimization issues. In particular for high-dimensional and structurally complex solution spaces, a proper balance between exploration, exploitation and fine-tuning is particularly important. This harmonization of these three phases is crucial to the algorithm working.

In the initial phase of the algorithm, Exploration, a GWO is used for searching the solution space in exhaustive manner within massive of solution space. In this phase, regions with reasonable potential solution identified and guarantee diversity in solution space. The GWO is not going to be stagnated in local optima by its global search capabilities and this provides a good groundwork to the later stages.

Now in the second stage, WOA comes in. In this stage more accurate search are performed into the promising areas found by GWO. WOA makes the global to local search swap with ease as it smoothly takes the candidate solutions discovered in the former phase to be made more accurate. In this refinement, the solution quality is improved by focusing on a more reduced area of search space.

The last one is fine-tuning with respect to AOA. Here, the best solutions found in the previous stages are subjected to detailed optimization processes in this stage. The local search capabilities of AOA allow the local detail in the solution to be investigated and the precision demanded is satisfied.

This three-tier structure enables easy integration of global search, local search and fine-tuning. Equilibrium between exploitation and exploration enables the efficient screening of solution space by obtaining high-precision results. Thus, this hybrid approach offers a robust and versatile solution for complex optimization problems.

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The proposed GWO_WOA_AOA framework is highly successful, depending on synchronized running of three main phases in a defined order. Each phases exploits the strengths of one algorithm to the next, which means that the optimization results are better than can be achieved by any single method alone. This structured integration ensures that the framework performs a thorough investigation on the large solution space and also conducts an accurate candidate solution refinement.

1) Working Principle of Hybrid GWO-WOA-AOA Optimization Algorithm

The working principle of three-stage hybrid optimization method in this division is discussed in detail. The algorithm is iterated through exploration (GWO), exploitation (WOA) and fine-tuning (AOA) phases with this combination, it is able to converge faster on the global optimum.

1. Exploration Phase: Grey Wolf Optimizer (GWO)

In the first phase of hybrid Algorithm, the GWO algorithm was adopted to effectively explore the large solution space. The following steps are included in this part, such as:

1.1. Initial Population

Between the lower (lb) and upper (ub) bounds, a random initial population of 20 wolves is generated according to equation (16). This large initial population allows efficient exploration of different regions of the solution space.

1.2. Control Parameter

Referring to equation (8), a, the main control parameter of the GWO, initially takes the value 2 ($a_{\max} = 2$) to encourage exploration behavior and gradually decreases towards 0 ($a_{\min} = 0$). This reduction allows the algorithm to transition from exploration to exploitation over 100 iterations gradually.

1.3. Creating a Social Hierarchy

In each iteration, all wolves' fitness values (objective function values) are calculated. Wolves are ranked according to their fitness value, and the top three wolves are designated alpha, beta, and delta, respectively. This hierarchy guides wolves' position updates.

1.4. Position Update

For each wolf, the coefficient vectors A_1, A_2, A_3 according to equation (3) and the coefficient vectors C_1, C_2, C_3 according to equation (4) are calculated. According to Equation (5), distance vectors representing the distances D_α, D_β and D_δ are calculated. Each wolf calculates X_1, X_2, and X_3 positions according to the positions of alpha, beta, and delta wolves according to equation (6).

1.5. Saving the Best Solution

The best solution obtained in each iteration is updated, and the best solution is passed to the next stage at the end of the algorithm.

This phase provides a strong exploration property, mainly thanks to the initially high parameter a. When $|A| > 1$, according to equation (3), wolves move away from the current prey and discover new potential solution regions.

2. Exploitation Phase: Whale Optimization Algorithm (WOA)

The best solution from the GWO is placed as the first whale of the WOA. A population of 20 whales is randomly generated between the lower (lb) and upper (ub) limits. Including the best solution of the previous phase allows the algorithm to move from a good starting point.

2.2. Control Parameters

In WOA, the parameter a decreases gradually from 2 to 0 according to equation (11). The spiral shape constant is set to $b = 1$ and defines the form of the logarithmic spiral movement in equation (13).

2.3. Position Updating Strategies

For each whale, a random $p - value$ is generated and compared to a threshold of 0.5. The coefficient vector A is computed employing equation (11), and the coefficient vector C is computed employing equation (12).

2.3.1. Prey Tracking ($p < 0.5$)

If $|A| < 1$, the whale moves towards the best solution according to Equations (9) and (10) by the "shrinking encircling" mechanism. If $|A| \geq 1$, the whale explores towards a randomly selected whale, moving according to Equations (14) and (15).

2.3.2. Spiral Updating ($p \geq 0.5$)

The whale performs a local search in a logarithmic spiral around the best solution according to equation (13). For this mechanism, the logarithmic spiral constant $b = 1$ is used, and the spiral motion is controlled by a randomly generated parameter l between -1 and 1.

2.4. Saving the Best Solution

In each iteration, if a better solution is found, the best solution is updated, and the best solution is passed to the next stage at the end of the algorithm.

At this stage, the WOA's "shrinking encircling" mechanism and spiral update strategy enable a more focused search in the promising region identified by the GWO.

3. Fine Tuning Phase: Arithmetic Optimization Algorithm (AOA

The third and final stage is fine-tuning using the Arithmetic Optimization Algorithm (AOA) to improve the solution from the WOA further. This stage consists of the following steps:

3.1. Initial Population

First individual of the AOA is the best solution from the WOA. 20 agents are created in the total population. Using the best value from the ended phase as predecessor allows the algorithme to continue from a good initiation point.

3.2. Control Parameters

A small value of $alpha = 0.1$ was chosen as an important parameter for fine-tuning the AOA. This smaller alpha value allows the MOP (Mathematical Operator Proportion) value in equation (17) to decrease more slowly, allowing the algorithm to fine-tune for longer. The parameter $mu = 0.499$ controls the magnitude of the changes to the

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solution space, and a value close to the limit is chosen to enable fine-tuning.

3.3. MOP Calculation

The MOP value is calculated at each iteration according to equation (17). The small $alpha$ value (0.1) allows the MOP to decrease slowly, allowing the algorithm to stay longer in the fine-tuning phase.

3.4. Position Updating Strategies

For each agent, a random r_1 value is generated and compared with the MOP:

3.4.1. Local Search Phase ($r_1 > MOP$)

The agent's location is refreshed with the local search strategy in this situation. If $r_3 < 0.5$, update according to equation (18). If $r_3 \geq 0.5$, it is updated according to equation (19). This mechanism allows for a more intensive investigation of the close surroundings of the prime solution.

3.4.2. Fine Tuning Phase ($r_1 \leq MOP$)

In this situation, the agent's location is refreshed with the fine-tuning strategy: If $r_2 < 0.5$, update according to equation (20). If $r_2 \geq 0.5$, the update is performed according to equation (21). This mechanism allows fine-tuning with minimal changes around the best solution.

3.5. Saving the Best Solution

In each iteration, the best solution and its value are updated if a better solution is found. At the end of the algorithm, the best solution and its value are returned as the final result of the hybrid algorithm.

In this stage, the small $alpha$ value (0.1) and the mu parameter (0.499) allow for micro-movements in the solution space, allowing fine-tuning of the solution found in the previous stages.

III. RESULTS

The Results of GWO_WOA_AOA was proven to be effective in the solution of complex optimization tasks and, therefore, validation of GWO_WOA_AOA was performed using extensive experiments in CEC2019 benchmark functions. As can be seen from the results of the test in Figure 1, the proposed hybrid method achieved superior results than other algorithms on most test functions done with respect to the CEC2019 benchmark suite. This benchmark has been proposed during the IEEE Congress on Evolutionary Computation 2019 and is a reliable and general testing ground for optimization algorithms [25]. It contains a portfolio of intricate functions that enhance the ability to measure whether algorithm can deal with multimodal, non-seperable and high dimensional problems fairly and majorly, compared to each other. The CEC2019 Test Functions in Table 1 are utilized to benchmark optimization algorithms and comprises a range of basic, mixed and hybrid shape functions [26].

TABLE 1 CEC 2019 Function Properties

No.	Function	D	Search Range	B
$\mathcal{F}_1$	Ackley Function	10	[−100, 100]	1
$\mathcal{F}_2$	Happy Cat Function	10	[−100, 100]	1
$\mathcal{F}_3$	Expanded Schaffer's F6 Function	10	[−100, 100]	1
$\mathcal{F}_4$	Modified Schwefel's Function	10	[−100, 100]	1
$\mathcal{F}_5$	Weierstrass Function	10	[−100, 100]	1
$\mathcal{F}_6$	Griewangk's Function	10	[−100, 100]	1
$\mathcal{F}_7$	Rastrigin's Function	10	[−100, 100]	1
$\mathcal{F}_8$	Lennard–Jones Minimum Energy Cluster	18	[−4, 4]	1
$\mathcal{F}_9$	Inverse Hilbert Matrix Problem	16	[−16,384, 16,384]	1
$\mathcal{F}_{10}$	Storn's Chebyshev Polynomial Fitting Problem	9	[–8192, 8192]	1

In Table 2, the control parameters of GWO, WOA, and AOA algorithms used in the proposed hybrid optimization method and the fundamental parameter values of Ant Lion Optimization (ALO), PSO, and DE techniques selected for comparison are presented in detail. To maintain fairness in the comparative framework, the population size (ps=20) and run cycles were kept constant at 50 for all algorithms. In the hybrid method, the maximum number of iterations of each algorithm is 100, and 300 iterations were performed. In order to evaluate the other methods under the same conditions, 300 iterations were carried out, and the parameter values that control the specific search strategy of each algorithm were chosen within the ranges commonly used in the literature.

TABLE 2 Control parameters employed in optimization approaches

Algorithm		Parameter	Value
GWO_ WOA_ AOA	GWO	ps, mi, a, A, C	20,100, 2→0, [-a, a], [0,2]
	WOA	ps, mi, a, b, A, C, p	20, 100, 2→0, 1, [-a, a], [0,2], [0,1]
	AOA	ps, mi, α, μ, r_1, r_2, r_3	20, 100, 0.1, 0.499, [0,1], [0,1], [0,1]
GWO		ps, mi, a, A, C	$20, 300, 2 \rightarrow 0, [-a, a], [0,2]$
WOA		ps, mi, a, A, C, P	$20,300, 2 \rightarrow 0, [-a, a], [0,2], [0,1]$
ALO		ps, mi, eae	$20, 300, 0.1$
PSO		ps, mi, w, c_1, c_2	$20, 300, 0.5, 1.5, 1.5$
DE		ps, mi, F, C_R	$20, 300, 0.5, 0.9$

Figure 1 illustrates the comparative performance results of the suggested GWO_WOA_AOA hybrid optimization method with the widely used DE, PSO, and ALO algorithms in the literature and the components of the hybrid method, GWO and WOA algorithms, on the CEC 2019 test functions. The outcomes demonstrate that the suggested GWO_WOA_AOA hybrid algorithm outperforms other optimization algorithms. The algorithm's effectiveness is measured using three distinct metrics: best values, average values, and standard deviation values.

When the best values in Figure 1 are examined, the GWO_WOA_AOA hybrid algorithm exhibits superior performance in many test functions. Especially in the Chebyshev function, it reached the optimal result with the value of 1.000e+00 and obtained better results compared to the other algorithms (GWO: 6.317e+02, WOA: 1.011e+00). In the Lennard-Jones function, it gave the best result among all algorithms with the value of 1.035e+00 (GWO: 1.706e+00, WOA: 4.285e+00). Similarly, in the Escaffer6 function, it performed better with the value of 3.452e+00 than GWO's 3.522e+00 and WOA's 4.094e+00. In the Rastrigin function, the value of 8.778e+01 was found to be superior to the values of GWO, 9.961e+01, and WOA, 1.763e+04.

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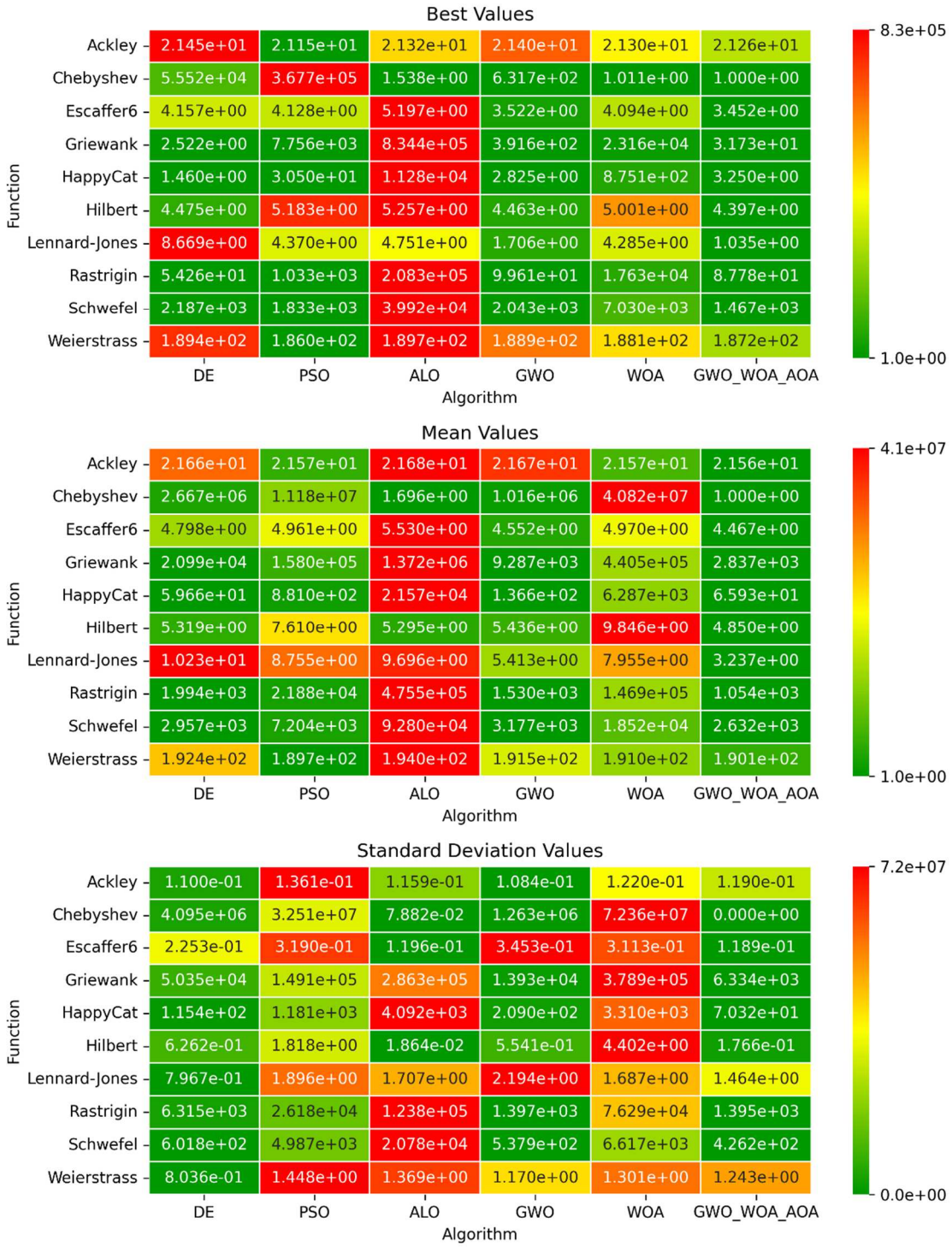


Fig.1. Performance comparison of the proposed hybrid method with existed method

When analyzed on average values, GWO_WOA_AOA hybrid algorithm shows a consistent performance. In Chebyshev function, it reaches an excellent result of 1.000e+00, while GWO remains at 1.016e+06 and WOA at 4.082e+07. In Lennard-Jones function, it achieves 3.237e+00, which is better than GWO's 5.413e+00 and WOA's 7.955e+00. In Griewank function, it performs much better with 2.837e+03, which is better than GWO's 9.287e+03 and WOA's 4.405e+05. In the Rastrigin function, it achieved superior results with a value of 1.054e+03 than GWO's 1.530e+03 and WOA's 1.469e+05. In the Weierstrass function, it reached the best average value among all algorithms with a value of 1.901e+02.

Standard deviation values are critical for the reliability and consistency of the algorithm. GWO_WOA_AOA hybrid algorithm exhibited excellent consistency with the value of 0.000e+00 in the Chebyshev function. This shows that the algorithm produces the same optimal result in repeated runs. In the Escaffer6 function, the value of 1.189e-01 showed lower standard deviation than the values of 3.453e-01 for GWO and 3.113e-01 for WOA. In the Griewank function, the value of 6.334e+03 obtained much lower standard deviation than the values of 1.393e+04 for GWO and 3.789e+05 for

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WOA. In the HappyCat function, the value of 7.032e+01 produced more consistent results than the values of 2.090e+02 for GWO and 3.310e+03 for WOA.

The GWO_WOA_AOA hybrid algorithm significantly improves the performance of its individual algorithms. For example, in the Chebyshev function, the best values of GWO are 6.317e+02 and WOA are 1.011e+00, while the hybrid algorithm reaches the optimal value of 1.000e+00. In the Lennard-Jones function, the best values of GWO are 1.706e+00 and WOA are 4.285e+00, while the hybrid algorithm provides a better result with 1.035e+00. In the Rastrigin function, the best values of GWO are 9.961e+01 and WOA are 1.763e+04, while the hybrid algorithm provides the best result with 8.778e+01. Similar improvements are also seen in the mean values. While the Griewank function obtained the mean values of GWO 9.287e+03 and WOA 4.405e+05, the hybrid algorithm performed much better with 2.837e+03. The hybrid algorithm also produced more consistent results than the individual algorithms in terms of standard deviation values. For example, while the Rastrigin function had the standard deviation values of GWO 1.397e+03 and WOA 7.629e+04, the hybrid algorithm gave more consistent results with 1.395e+03.

These results show that the GWO_WOA_AOA hybrid algorithm successfully combines the advantages of GWO's exploration ability, WOA's exploitation ability, and AOA's mathematical operators, and significantly improves the performance of individual algorithms by creating a synergistic effect. The hybrid approach is capable of producing more effective, consistent, and reliable results in complex optimization problems.

IV. CONCLUSION

This paper presents a novel hybrid optimization methodology combining the GWO, WOA and AOA algorithms. In the three-phase structure of the proposed model, the GWO algorithm for exploration phase, WOA algorithm for exploitation phase and AOA algorithm for fine-tuning. Comprehensive empirical studies on CEC2019 benchmark functions have demonstrated that the hybrid approach provides better results than some conventional techniques. The algorithm exhibits faster convergence and superior quality solutions, as shown by the observations in particular for multidimensional and multimodal cases. The testing results show that the proposed hybrid technique leads to a significant improvement in the results as compared to when algorithms are executed individually (GWO and WOA). The results indicate that the three-stage hybrid architecture is a good way to solve complicated optimization issues.

Next future works will evaluate the proposed hybrid approach to real problems and study its combinations with other meta-heuristic.

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